

Dr. Dhruv Kumar

BIO-DATA

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Personal Details

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Name	Dr. Dhruv Kumar
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Academic Qualifications

#	Degree	Year	Subject	University / Institution	CGPA
1	B.E. (Hons.)	2014	Computer Science	Birla Institute of Technology and Science, Pilani	9.92/10
2	Ph.D.	2022	Computer Science & Engineering	University of Minnesota, Twin Cities, USA	4.0/4.0

Work Experience

#	Position	Institution	From	To
1	Assistant Professor	BITS Pilani	May 2024	Present
2	Adjunct Faculty	IIIT Delhi	May 2024	Present
3	Assistant Professor	IIIT Delhi	Aug 2022	May 2024
4	Post-Doctoral Researcher	Microsoft Research, Bengaluru	Mar 2022	Jul 2022
5	Ph.D. Intern	Google Cloud, California, USA	May 2019	Aug 2019
6	Software Engineer / Quant Analyst	Goldman Sachs, Bengaluru	Nov 2014	Apr 2016
7	Summer Intern	CSIR-CEERI, Pilani	May 2012	Jul 2012

Professional Recognition / Awards / Prizes / Fellowships

#	Award / Recognition	Awarding Body	Year
1	Teaching Excellence Award (AY 2023–24)	IIIT Delhi	Sep 2024
2	Teaching Excellence Award (AY 2022–23)	IIIT Delhi	Sep 2023
3	3M Science and Technology Fellowship	University of Minnesota, Twin Cities	2017–2021
4	Travel Grant	USENIX ATC	2019
5	Travel Grant	ACM SIGMETRICS	2019
6	Travel Grant	USENIX OSDI	2018
7	Merit Scholarship (Top 10 among 800 students)	BITS Pilani	2010–2014
8	Outstanding Undergraduate Thesis	BITS Pilani	2014

Publications (in Year-wise Descending Order)

Author's name in **bold**. [*Best Paper Award*] denotes prize-winning papers.

Journal Articles & Transactions

- J1.** Alakh Sharma, Gaurish Trivedi, Kartikey Singh Bhandari, Yash Sinha, **Dhruv Kumar**, Pratik Narang, and Jagat Sesh Challa. 2026. Generative Evolutionary Meta-Solver (GEMS): Scalable Surrogate-Free Multi-Agent Reinforcement Learning. **Transactions on Machine Learning Research (TMLR) 2026**.
- J2.** Raj Gupta, Harshita Goyal, **Dhruv Kumar**, Apurv Mehra, Sanchit Sharma, Kashish Mittal, and Jagat Sesh Challa. 2026. Saksham AI: Advancing AI-Assisted Coding Education for Engineering Students in India Through Socratic Tutoring and Comprehensive Feedback. **ACM Journal on Computing and Sustainable Societies (JCSS) 2026**.
- J3.** Harshita Goyal, Garima Garg, Prisha Mordia, Veena Ramachandran, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. The Impact of Large Language Models on K-12 Education in Rural India: A Thematic Analysis of Student Volunteers' Perspectives. **Scientific Reports (2025)**.
- J4.** Balamurugan Palanisamy, Vikas Hassija, Arpita Chatterjee, Arpita Mandal, Debanshi Chakraborty, Amit Pandey, G. Sai Seshu Chalapathi, and **Dhruv Kumar**. 2025. Transformers for Vision: A Survey on Innovative Methods for Computer Vision. **IEEE Access 13 (2025)**, 95496–95523.
- J5.** Harshal Akolekar, Piyush Jhamnani, Vikash Kumar, Vinay Tailor, Aditya Pote, Ankit Meena, Kamal Kumar, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. The Role of Generative AI Tools in Shaping Mechanical Engineering Education from an Undergraduate Perspective. **Scientific Reports 15 (2025)**, Article 9214.
- J6.** Ashutosh Sharma, **Dhruv Kumar**, Srinivas Rallapalli, et al. 2023. Wetland Functional Assessment and Uncertainty Analysis Using Fuzzy Alpha-Cut-Based Modified Hydrogeomorphic Approach. **Environmental Science & Pollution Research 30 (2023)**, 72900–72915.
- J7.** Ananya Jain, Srinivas Rallapalli, and **Dhruv Kumar**. 2022. Cloud-Based Neuro-Fuzzy Hydro-Climatic Model for Water Quality Assessment under Uncertainty and Sensitivity. **Environmental Science & Pollution Research 29 (2022)**, 65259–65275.
- J8.** Poonam Goyal, Jagat Sesh Challa, **Dhruv Kumar**, Anuvind Bhat, Sundar Balasubramaniam, and Navneet Goyal. 2020. Grid-R-Tree: A Data Structure for Efficient Neighborhood and Nearest Neighbor Queries in Data Mining. **Int. J. Data Sci. Anal. 10 (2020)**, 25–47.
- J9.** **Dhruv Kumar**, Jian Li, Abhishek Chandra, and Ramesh K. Sitaraman. 2019. A TTL-Based Approach for Data Aggregation in Geo-Distributed Streaming Analytics. **Proc. ACM MACS 3, 2 (2019)**, 29:1–29:27.
- J10.** **Dhruv Kumar**, Poonam Goyal, and Navneet Goyal. 2018. An Efficient Method for Batch Updates in OPTICS Cluster Ordering. **Int. J. Data Anal. Tech. Strateg. 10 (2018)**, 57–80.

Conference Papers

- C1.** Aaryaman Kalani, Murari Mandal, **Dhruv Kumar**, Mohan Kankanhalli, and Yash Sinha. 2026. The BatchNorm Illusion: Diagnosing Normalization Artifacts in Machine Unlearning Evaluation. In **NeurIPS 2026**.
- C2.** Stuti Sinha, Himanshu Kumar, Aryan Raju Mandapati, Rakshit Sakhuja, and **Dhruv Kumar**. 2026. The Language of Bargaining: Linguistic Effects in LLM Negotiations. In **AACL-IJCNLP 2026 (Main Conference)**, also accepted

at **ICLR 2026 Workshop (LLM Reasoning)**, **ACL 2026 Workshop (C3NLP)**, **ICML 2026 Workshop (Culture x AI)**, and **ICML 2026 Workshop (Pluralistic Alignment)**.

- C3.** Darshil Chauhan, Adityasinh Solanki, Vansh Patel, Kanav Kapoor, Ritvik Jain, Aditya Bansal, **Dhruv Kumar**, and Pratik Narang. 2026. Navigating the Reality Gap: Privacy-Preserving On-Device Continual Adaptation of ASR for Clinical Telephony. In **AACL-IJCNLP 2026 (Main Conference)**.
- C4.** Dhruv Gupta, Vasudev Majhi, Jagat Sesh Challa, Murari Mandal, and **Dhruv Kumar**. 2026. Anchored Turns: A Length-Driven Gradient in Chain-of-Thought Causal Necessity Across Multi-Turn Conversations. In **AACL-IJCNLP 2026 (Main Conference)**.
- C5.** Venkat Aniruddh Thammatam, Vasudev Majhi, Aaryan Nagda, Jagat Sesh Challa, Murari Mandal, Yash Sinha, and **Dhruv Kumar**. 2026. Method Ranking Inverts Across Targets: A Matched-Grid Evaluation of Jailbreak Attacks on Open-Weight Vision-Language Models. In **AACL-IJCNLP 2026 (Main Conference)**.
- C6.** Devanshu Sahoo, Manish Prasad, Vasudev Majhi, Jahnvi Singh, Vinay Chamola, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2026. When Reject Turns into Accept: Quantifying the Vulnerability of LLM-Based Scientific Reviewers to Indirect Prompt Injection. In **Findings of AACL-IJCNLP 2026**.
- C7.** Shiven Sharma, Gautam Choudhary, Ashish Sam, and **Dhruv Kumar**. 2026. Benchmarking Frontier Agentic Deception in LLMs vs Humans. In **Findings of AACL-IJCNLP 2026**.
- C8.** Debdeep Sanyal, Aakash Sen Sharma, **Dhruv Kumar**, Saurabh Deshpande, and Murari Mandal. 2026. Format Collapse in Policy Optimization for LLM Reasoning. In **EMNLP 2026**.
- C9.** Soham Roy, Sarthakbrata Halder, Arya Bharaty, Vaibhav Bhaskar, Yash Sinha, **Dhruv Kumar**, Srikant Panda, and Murari Mandal. 2026. “I Strongly Suspect This Website Is a Scam”: Benchmarking PII Leakage and Detection without Defense in Autonomous Web Agents. In **EMNLP 2026**.
- C10.** Debdeep Sanyal, Umakanta Maharana, Hong Ming Tan, Yash Sinha, Shirish Karande, Mohan Kankanhalli, **Dhruv Kumar**, and Murari Mandal. 2026. OrgAccess: A Benchmark for Role-Based Access Control in Organization Scale LLMs. In **EMNLP 2026**.
- C11.** Sasank Tumpati, Devansh Agarwal, Ayush Kedia, Arjun Neekhara, Murari Mandal, Krishna Garg, Yash Sinha, Suman Gupta, and **Dhruv Kumar**. 2026. FinBalance: A Multi-Document Accounting Reconciliation Benchmark. In **Findings of EMNLP 2026**.
- C12.** Aniket Upadhyay, Aakash Trivedi, Sparsh Bajoria, Shreyanshu Ranjan, **Dhruv Kumar**, and Pratik Narang. 2026. PRAXIS: A 68-Industry Benchmark for Pragmatic-Form Bias in Suggestion Mining. In **Findings of EMNLP 2026**.
- C13.** Aniket Upadhyay, Aakash Trivedi, **Dhruv Kumar**, and Pratik Narang. 2026. Certified Cost-Optimal Triage for Suggestion Detection in Reviews. In **EMNLP 2026 Industry Track**.
- C14.** Tulika Tewari, Nalin A. G. Arachchilage, Jagat Sesh Challa, and **Dhruv Kumar**. 2026. From Trust to Compromise: Outcome-Verified LLM Phishing Simulation and Real-Time Defense. In **Association for Computational Linguistics (ACL 2026)**.
- C15.** Samyak Savi, Chavi Gupta, Shreyas Gantayet, Tanay Sodha, and **Dhruv Kumar**. 2026. Evaluating and Mitigating Misgendering in English-to-Hindi Machine Translation. In **COLM 2026**, also accepted at **ICML 2026 Workshop (Culture x AI)**.
- C16.** **Dhruv Kumar**. 2026. Oracle-Agnostic Text Quality Optimization via Prompt-Only Agentic Loops. In **CAISc 2026**.
- C17.** Vikranth Udandaraao, Noel Abraham Tiju, Muthuraj Vairamuthu, Harsh Mistry, and **Dhruv Kumar**. 2026. Roamify: Designing and Evaluating an LLM-Based Google Chrome Extension for Personalised Itinerary Planning. In **The Web Conference (WWW) 2026**.
- C18.** Pankaj Iitd, **Dhruv Kumar**, Vinayak Abrol, and Vikram Goyal. 2026. BLEND: Balanced and Leaf-Enhanced Dual Fine-Tuning for Taxonomy Completion. In **The Web Conference (WWW) 2026**.
- C19.** Aakash Trivedi, Aniket Upadhyay, Pratik Narang, **Dhruv Kumar**, and Praveen Kumar. 2026. A Hybrid Supervised-LLM Pipeline for Actionable Suggestion Mining in Unstructured Customer Reviews. In **EACL 2026 Industry Track**.
- C20.** Debdeep Sanyal, Agniva Maiti, Umakanta Maharana, **Dhruv Kumar**, Ankur Mali, C. Lee Giles, and Murari Mandal. 2025. Investigating Pedagogical Teacher and Student LLM Agents: Genetic Adaptation Meets Retrieval-Augmented Generation Across Learning Styles. In **EMNLP 2025**, 13359–13400.
- C21.** Varun Bharti, Shashwat Jha, **Dhruv Kumar**, and Pankaj Jalote. 2025. Combining Reasoning Optimised LLMs and SMT Solvers for Automated Loop Invariant Synthesis. In **AIWare 2025**. **[Best Paper Award]**
- C22.** Aditya Pathak, Rachit Gandhi, Vaibhav Uttam, Arnav Ramamoorthy, Pratyush Ghosh, Aaryan Raj Jindal, Shreyash Verma, Aditya Mittal, Aashna Ased, Chirag Khatri, Yashwanth Nakka, Devansh, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. Rubric Is All You Need: Improving LLM-Based Code Evaluation With Question-Specific Rubrics. In **ICER 2025**, 181–195.
- C23.** Adithya Chittem, Aishna Shrivastava, Sai Tarun Pendela, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. SAC: A Framework for Measuring and Inducing Personality Traits in LLMs with Dynamic Intensity Control. In **ICAART 2025**.
- C24.** Madhav Krishan Garg, Tejash Prasad, Tanmay Singhal, Chhavi Kirtani, Murari Mandal, and **Dhruv Kumar**. 2025. ReviewEval: An Evaluation Framework for AI-Generated Reviews. In **Findings of EMNLP 2025**, 20542–20564.
- C25.** Utkarsh Arora, Anupam Garg, Aryan Gupta, Samyak Jain, Ronit Mehta, Rupin Oberoi, Prachi, Aryaman Raina,

Manav Saini, Sachin Sharma, Jaskaran Singh, Sarthak Tyagi, and **Dhruv Kumar**. 2025. Analyzing LLM Usage in an Advanced Computing Class in India. In **ACE '25**, 154–163.

- C26.** Ritvik Budhiraja, Ishika Joshi, Jagat Sesh Challa, Harshal D. Akolekar, and **Dhruv Kumar**. 2024. “It’s not like Jarvis, but it’s pretty close!”—Examining ChatGPT’s Usage among Undergraduate Students in CS. In **ACE 2024**, 124–133.
- C27.** Shubhang Shekhar Dvivedi, Vyshnav Vijay, Sai Leela Rahul Pujari, Shoumik Lodh, and **Dhruv Kumar**. 2024. A Comparative Analysis of Large Language Models for Code Documentation Generation. In **AIware 2024**.
- C28.** Harshal Dev, Raj Gupta, and **Dhruv Kumar**. 2024. From Cash to Cashless: UPI’s Impact on Spending Behavior among Indian Users. In **CHI EA 2024**, 168:1–168:10.
- C29.** Shikhar Sharma, Manas Mhasakar, Apurv Mehra, Utkarsh Venaik, Ujjwal Singhal, **Dhruv Kumar**, and Kashish Mittal. 2024. Comuniqa: Exploring Large Language Models For Improving English Speaking Skills. In **COMPASS 2024**, 256–267.
- C30.** Ishika Joshi, Ritvik Budhiraja, Harshal Dev, Jahnvi Kadia, Mohammad Osama Ataullah, Sayan Mitra, Harshal D. Akolekar, and **Dhruv Kumar**. 2024. ChatGPT in the Classroom: An Analysis of Its Strengths and Weaknesses for Solving Undergraduate CS Questions. In **SIGCSE 2024**, 625–631.
- C31.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2023. AggFirstJoin: Optimizing Geo-Distributed Joins using Aggregation-Based Transformations. In **CCGrid 2023**, 414–425. **[Best Paper Award]**
- C32.** Vikram Goyal, Naveen Kumar, Sourav S. Bhowmick, Pawan Goyal, Navneet Goyal, and **Dhruv Kumar** (Eds.). 2023. Big Data and Artificial Intelligence—11th International Conference, **BDA 2023**, Proceedings. Lecture Notes in Computer Science 14418. Springer.
- C33.** Joel Wolfrath, Nikhil Sreekumar, **Dhruv Kumar**, Yuanli Wang, and Abhishek Chandra. 2022. HACCS: Heterogeneity-Aware Clustered Client Selection for Accelerated Federated Learning. In **IPDPS 2022**, 985–995.
- C34.** Rallapalli Srinivas, Matt Drewitz, Joe Magner, Ajit Pratap Singh, **Dhruv Kumar**, and Yashwant Bhaskar Katpatal. 2022. Simulating Landscape Hydrologic Connectivity in a Precise Manner Using Hydro-Conditioning. In **Advances in Computational Modeling and Simulation**, Lecture Notes in Mechanical Engineering. Springer.
- C35.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2021. AggNet: Cost-Aware Aggregation Networks for Geo-Distributed Streaming Analytics. In **SEC 2021**, 297–311.
- C36.** Sonal Kumari, Poonam Goyal, Ankit Sood, **Dhruv Kumar**, Sundar Balasubramaniam, and Navneet Goyal. 2017. Exact, Fast and Scalable Parallel DBSCAN for Commodity Platforms. In **ICDCN 2017**, Article 14.
- C37.** Poonam Goyal, Sonal Kumari, Sumit Sharma, **Dhruv Kumar**, Vivek Kishore, Sundar Balasubramaniam, and Navneet Goyal. 2016. A Fast, Scalable SLINK Algorithm for Commodity Cluster Computing Exploiting Spatial Locality. In **HPCC 2016**, 268–275.
- C38.** Poonam Goyal, Sonal Kumari, **Dhruv Kumar**, Sundar Balasubramaniam, Navneet Goyal, Saiyedul Islam, and Jagat Sesh Challa. 2015. Parallelizing OPTICS for Commodity Clusters. In **ICDCN 2015**, 33:1–33:10.
- C39.** Poonam Goyal, Sonal Kumari, **Dhruv Kumar**, Sundar Balasubramaniam, and Navneet Goyal. 2014. Parallelizing OPTICS for Multicore Systems. In **ACM COMPUTE 2014**, 17:1–17:6.

Workshop Papers, Posters & Abstracts

- W1.** Aviral Gupta, Armaan Sethi, and **Dhruv Kumar**. 2026. TabPFN Through The Looking Glass: An Interpretability Study of TabPFN and Its Internal Representations. In **NeurIPS 2026 Workshop (XAI4Science)**.
- W2.** Shoyam Kumar Mishra and **Dhruv Kumar**. 2026. Persistent Data-Order Effects Precede Directional Alignment in Language Models. In **NeurIPS 2026 Workshop (DynaFront)**.
- W3.** Vaibhav Devraj, **Dhruv Kumar**, Jagat Sesh Challa, Parth Agarwal, Navya Kommuri, Trizal Garg, Prisha Singhal, Dhruv Shah, and Yash Sinha. 2026. CricBench: A Multilingual Benchmark for Evaluating LLMs in Cricket Analytics. In **EMNLP 2026 Workshop (REALM)**.
- W4.** Parth Agarwal, Sanchit Bahuguna, and **Dhruv Kumar**. 2026. LLM Graders Are Systematically Harsher Than Industry Interviewers: A Benchmark of 221 Real Technical Interviews. In **EMNLP 2026 Workshop (REALM)**.
- W5.** Yash Sinha, Parth Shrikant Patil, Murari Mandal, and **Dhruv Kumar**. 2026. Beyond Accuracy: Diagnosing Algebraic Reasoning Failures in LLMs Across Nine Complexity Dimensions. In **EMNLP 2026 Workshop (MathNLP)**.
- W6.** Vedant Jawandhia, Yash Sinha, Ankan Pal, Murari Mandal, and **Dhruv Kumar**. 2026. Measuring Representation Robustness in Large Language Models for Geometry. In **ICML 2026 Workshop (AI4Math)**.
- W7.** Manan Gupta, Chinmay Pushkar, Sanchit Kabra, **Dhruv Kumar**, and Jagat Sesh Challa. 2026. MultiVulnBench: A Large-Scale Benchmark for Count Bias in LLM-Based Multi-Vulnerability Detection. In **ICML 2026 Workshop (CTB)**, also accepted at **AgenticUQ**, **CompLearn**, **FAGEN**, and **DL4C**.
- W8.** Ojaswi Prakash, **Dhruv Kumar**, Murari Mandal, Mohan Kankanhalli, and Yash Sinha. 2026. MafiaPersona: A Multi-Agent Adversarial Benchmark for Evaluating Persona Persistence in Large Language Models. In **ICML 2026 Workshop (NExT-Game)**.
- W9.** Prapti Patra and **Dhruv Kumar**. 2026. PoolBench: Benchmarking Large Language Models on Continuous Physical Action Selection in Eight-Ball Pool. In **ICML 2026 Workshop (NExT-Game)**.
- W10.** Preetham Sivalingam, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. TSQueryBench: LLM-as-a-Judge for Time-Series Explanations. In **ICML 2026 Workshop (FMSD)**.
- W11.** Moulik Gupta, Aaditya Jain, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. ReGeN: Reference-

Guided Synthetic Multivariate Time Series Generation for Forecasting. In **ICML 2026 Workshop (FMSD)**.

- W12.** Atharva Gupta, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. Where Computation Lives Inside TabPFN: Causal Localisation of Attention Head Function. In **ICML 2026 Workshop (FMSD)**.
- W13.** Jay Agarwal, Siddharth Khare, and **Dhruv Kumar**. 2026. What Do Lorentz-Equivariant Jet Taggers Learn? In **ICML 2026 Workshop (AI4Physics)**.
- W14.** Shivansh Bibra, **Dhruv Kumar**, Murari Mandal, and Yash Sinha. 2026. Joint Evaluation of Compliance, Planning, and Consistency under Paraphrase: A Relational-Complexity View of Frontier LLMs. In **ICML 2026 Workshop (CTB)**.
- W15.** Jalluri Mahesh Kumar, Junjunoori Sri Chakri, Yash Kothari, Murari Mandal, Yash Sinha, and **Dhruv Kumar**. 2026. Evaluating LLM Reasoning on Operating System Algorithms via Step-Level Verification. In **ICML 2026 Workshop (CTB)**.
- W16.** Kanav Kapoor, Jagat Sesh Challa, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2026. GraphStateEval: A Step-by-Step Evaluation Framework for Graph Algorithm Execution in Large Language Models via Intermediate State Tracing. In **ICML 2026 Workshop (CTB)**.
- W17.** Shubh Chapra, **Dhruv Kumar**, Murari Mandal, and Yash Sinha. 2026. The Complexity Ceiling Benchmark: A Multi-Domain Evaluation of Sequential Reasoning Under Depth Scaling. In **ICML 2026 Workshop (CTB)**.
- W18.** Aryan Khurana, Aravind Ramana RN, and **Dhruv Kumar**. 2026. Authority, Truth, and Citation Bias: A Large-Scale Multi-Domain Benchmark for Studying Epistemic Susceptibility in Large Language Models. In **ICML 2026 Workshop (EIML)**, also accepted at **AI4GOOD** and **NeurIPS 2026 Workshop (AI4GOOD)**.
- W19.** Agniva Maiti, Aditya Tiwari, Dwarikanath Mahapatra, Sudipta Roy, Yash Sinha, **Dhruv Kumar**, and Murari Mandal. 2026. Bridging the Visual Gap: Resource-Efficient VLM Adaptation for Meitei Mayek using Synthetic Multi-Modal Mixtures. In **ICML 2026 Workshop (GlobalSouthML)**.
- W20.** Agniva Maiti, Aarsh Muley, Aditya Tiwari, Dwarikanath Mahapatra, Sudipta Roy, Yash Sinha, **Dhruv Kumar**, and Murari Mandal. 2026. Evaluating Indic Literary Creativity In LLMs through the Lens of the Shastras. In **ICML 2026 Workshop (GlobalSouthML)**.
- W21.** Aviral Dawar, Roshan Karanth, Vikram Goyal, and **Dhruv Kumar**. 2026. IndicDB: Benchmarking Multilingual Text-to-SQL Capabilities in Indian Languages. In **ICML 2026 Workshop (Culture x AI)**, also accepted at **GlobalSouthML** and **ACL 2026 Workshop (MeLLM)**.
- W22.** Manya Pandey, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. GITCO: Gated Inference-Time Context Optimization in TSFMs. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD**.
- W23.** Aaryan Nagpal, Debdeep Sanyal, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. Mix, Don't Pick: Why Synthetic Corpus Composition Matters for Time Series Foundation Model Pretraining. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD** and **Forecast**.
- W24.** Karthik Sridhar, Atharva Gupta, Nishant Pradhan, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. Semantics or Structure? Auditing Text Sensitivity in Multimodal Time-Series Forecasting. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD**.
- W25.** Manan Gupta and **Dhruv Kumar**. 2026. Diagnosing LLM Judge Reliability: Conformal Prediction Sets and Transitivity Violations. In **ICML 2026 Workshop (SCALE)**, also accepted at **CTB**, **SPIGM**, **FoGen**, and **FAGEN**.
- W26.** Manan Gupta and **Dhruv Kumar**. 2026. Is the Trust Region Necessary? Robust Entropy Adaptation Suffices for Continuous Control. In **ICML 2026 Workshop (SCALE)**.
- W27.** Manan Gupta and **Dhruv Kumar**. 2026. Latent Phase-Shift Rollback: Inference-Time Error Correction via Residual Stream Monitoring and KV-Cache Steering. In **ICML 2026 Workshop (SCALE)**, also accepted at **AgenticUQ**, **CTB**, **FoGen**, and **FAGEN**.
- W28.** Manan Gupta and **Dhruv Kumar**. 2026. Reasoning Phases Are Continuous, Not Discrete: Evidence from Switching Linear Dynamical Systems Applied to Chain-of-Thought Residual Streams. In **ICML 2026 Workshop (SCALE)**, also accepted at **CompLearn**, **FoGen**, and **Mech Interp**.
- W29.** Manan Gupta, Inderjeet Jayakumar Nair, Lu Wang, and **Dhruv Kumar**. 2026. Context Over Content: Exposing Evaluation Faking in Automated Judges. In **ICML 2026 Workshop (AgenticUQ)**, also accepted at **SPIGM**, **FoGen**, **FAGEN**, and **AI4GOOD**.
- W30.** Manan Gupta and **Dhruv Kumar**. 2026. DPMI: A Principled Index for Neural Polysemanticity via Dirichlet Process Mixture Modeling. In **ICML 2026 Workshop (SCALE)**, also accepted at **CompLearn** and **FoGen**.
- W31.** Aaryaman Kalani, **Dhruv Kumar**, Mohan Kankanhalli, Murari Mandal, and Yash Sinha. 2026. RASE: Retain-Agnostic Machine Unlearning via Activation Subspace Projection. In **CVPR 2026 Workshop (MUV)**.
- W32.** Shubham Mishra, Shiv Tiwari, Samyek Jain, Gorang Mehrishi, **Dhruv Kumar**, Pratik Narang, and Harsh Sharma. 2026. From Facts to Conclusions: Integrating Deductive Reasoning in Retrieval-Augmented LLMs. In **ICLR 2026 Workshop on LLM Reasoning**. arXiv (2025).
- W33.** Debdeep Sanyal, Aaryan Nagpal, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2025. time2time: Causal Intervention in Hidden States to Simulate Rare Events in Time Series Foundation Models. In **NeurIPS 2025 Workshop on Time Series Foundation Models (BERT4S)**.
- W34.** Sankalp Tattwadarsai Swain, Anshika Krishnatray, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. Talking with Oompa Loompas: A Novel Framework for Evaluating Linguistic Acquisition of LLM Agents. In **NeurIPS 2025**

Workshop (WiML), also accepted at **LLM Evaluation**.

- W35.** Utkarsh Tiwari, Aryan Seth, Adi Mukherjee, Kaavya Mer, Kavish, and **Dhruv Kumar**. 2025. DebateBench: A Challenging Long Context Reasoning Benchmark For Large Language Models. In **NAACL SRW 2025**.
- W36.** Shreya Bhatia, Tarushi Gandhi, **Dhruv Kumar**, and Pankaj Jalote. 2024. Unit Test Generation using Generative AI: A Comparative Performance Analysis of Autogeneration Tools. In **LLM4Code@ICSE 2024**, 54–61.
- W37.** **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2023. AggFirstJoin (Workshop Version). In **CCGridW 2023**, 308–310.
- W38.** **Dhruv Kumar**, Joel Wolfrath, Abhishek Chandra, and Ramesh K. Sitaraman. 2022. Towards WAN-Aware Join Sampling over Geo-Distributed Data. In **EdgeSys@EuroSys 2022**, 13–18.
- W39.** Yuanli Wang, Joel Wolfrath, Nikhil Sreekumar, **Dhruv Kumar**, and Abhishek Chandra. 2021. Accelerated Training via Device Similarity in Federated Learning. In **EdgeSys@EuroSys 2021**, 31–36.
- W40.** Yuanli Wang, **Dhruv Kumar**, and Abhishek Chandra. 2020. Exploiting Data Heterogeneity for Performance and Reliability in Federated Learning (Poster). In **SEC 2020**, 164–166.
- W41.** **Dhruv Kumar**, Aravind Alagiri Ramkumar, Rohit Sindhu, and Abhishek Chandra. 2019. DeCaf: Iterative Collaborative Processing over the Edge. In **HotEdge 2019**.
- W42.** **Dhruv Kumar**, Jian Li, Abhishek Chandra, and Ramesh K. Sitaraman. 2019. A TTL-Based Approach for Data Aggregation in Geo-Distributed Streaming Analytics (Abstract). In **SIGMETRICS 2019**, 15–16.

Preprints (arXiv / SSRN)

- P1.** Rakshita Goel, S. Pranav Kumar, Anmol Agrawal, Divyan Poddar, Pratik Narang, and **Dhruv Kumar**. 2026. Domain-Partitioned Hybrid RAG for Legal Reasoning: Toward Modular and Explainable Legal AI for India. **arXiv (2026)**.
- P2.** Gaurish Trivedi, Alakh Sharma, Kartikey Singh Bhandari, Yash Sinha, Pratik Narang, **Dhruv Kumar**, and Jagat Sesh Challa. 2026. Trust Regions Sell, But Who’s Buying? Overlap Geometry as an Alternative Trust Region for Policy Optimization. **arXiv (2026)**.
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- P28.** Ishika Joshi, Ritvik Budhiraja, Pranav Deepak Tanna, Lovenya Jain, Mihika Deshpande, Arjun Srivastava, Srinivas Rallapalli, Harshal D. Akolekar, Jagat Sesh Challa, and **Dhruv Kumar**. 2023. "With Great Power Comes Great Responsibility!": Student and Instructor Perspectives on the Influence of LLMs on Undergraduate Engineering Education. **arXiv:2309.10694 (2023)**.
- P29.** Atharva Mehta, Nipun Gupta, Aarav Balachandran, **Dhruv Kumar**, Pankaj Jalote, et al. 2023. Can ChatGPT Play the Role of a Teaching Assistant in an Introductory Programming Course? **arXiv (2023)**.

Sponsored Research Projects

1. SPANDAN — AI/ML for Strengthening Police Investigation Processes	
Funding Agency	Gujarat Police (Office of Superintendent of Police, Ahmedabad Rural)
Budget	INR 34,80,000
Duration	11 months (03/2026–01/2027)
Status	Ongoing
Role	PI

2. Revolutionizing Indian Higher Education Institutes with Generative AI	
Funding Agency	BITS Pilani
Budget	INR 20,00,000
Duration	24 months (11/2024–10/2026)
Status	Ongoing
Role	PI

Summary of Results & Publications:

- Pathak, A.; **Kumar, D.**; et al. 2025. Rubric Is All You Need: LLM-based code evaluation. *ACM ICER 2025*.
- Arora, U.; **Kumar, D.**; et al. 2025. Analyzing LLM usage in an advanced computing class in India. *ACE '25*, 154–163.

3. AI for River–Climate Modeling	
Funding Agency	The Nature Conservancy (NGO)
Budget	INR 25,00,000
Duration	11 months (08/2024–06/2025)
Status	Completed
Role	Co-PI

Summary of Results & Publications:

- Rallapalli, S.; **Kumar, D.**; et al. 2025. Quantifying the role of land use–land cover for adaptive river basin management. *Journal of Environmental Management* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Targeting degraded hotspots in riparian corridors for rehabilitation. *Journal of Environmental Management* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Quantifying the synergistic influence of LULC and climate on hydro-climatological dynamics. *Journal of Hydrology* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Disaggregating the influence of climate change and LULC on streamflow. *Journal of Environmental Management* (under submission).

4. Generative AI in Education Grant	
Funding Agency	WISE–IIE Higher Education Consortium
Budget	INR 30,00,000
Duration	10 months (08/2024–05/2025)
Status	Completed
Role	Co-PI

Summary of Results & Publications:

- Akolekar, H.; **Kumar, D.**; et al. 2025. Generative AI tools in mechanical engineering education. *Scientific Reports* 15, 9214.
- Gupta, R.; **Kumar, D.**; et al. 2025. Saksham AI: Socratic tutoring for coding education in India. *ACM JCSS 2026*.
- Goyal, H.; **Kumar, D.**; et al. 2025. Impact of LLMs on K-12 education in rural India. *Scientific Reports 2025*.
- Sanyal, D.; **Kumar, D.**; et al. 2025. Investigating pedagogical teacher and student LLM agents. *EMNLP 2025*.

5. Google Cloud Education Grant	
Funding Agency	Google Cloud
Budget	INR 12,50,000
Duration	12 months (01/2024–12/2024)
Status	Completed
Role	PI